JGR
JJR

Ⓓ

Duration   Decemba – 1990

Why correlation

Decadal wise [No of grids] → Map
" " → Season
Total no of grids → JJA, OWT

Regional

① ← IMD Homogeneous region
Koppen climatic region

Region & Seasons   Monsoonal selection / Region [

Regional / Seasonal

⑬

.nc

panoply =

.netCDF

ncview

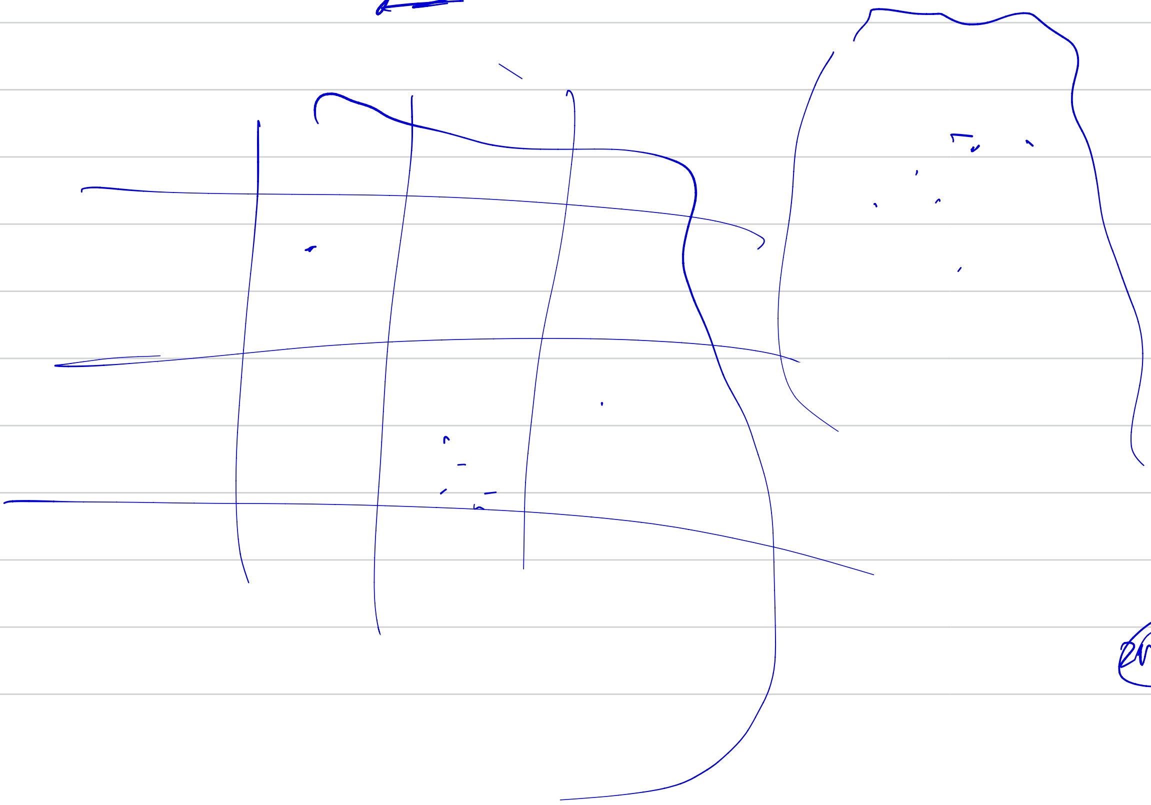
.grib

1) Observational dataset → 1951 - 2025

gridded

IMD

daily



>600 km²

200 km²

1x1

.25°x.25°

Min Temp

Max Temp

Precip.

1951-2025

daily

1951-2025

1901

Imd

M.Tech

Stats

② Reanalysis dataset

ERA5

NOAA

IMD

Model + Observ

Data Assimilation

Atmospheric

V → Note

Imc

Gene

1951

2025

25mm

35mm global

.25x.25

Hourly

Daic

1951

2021

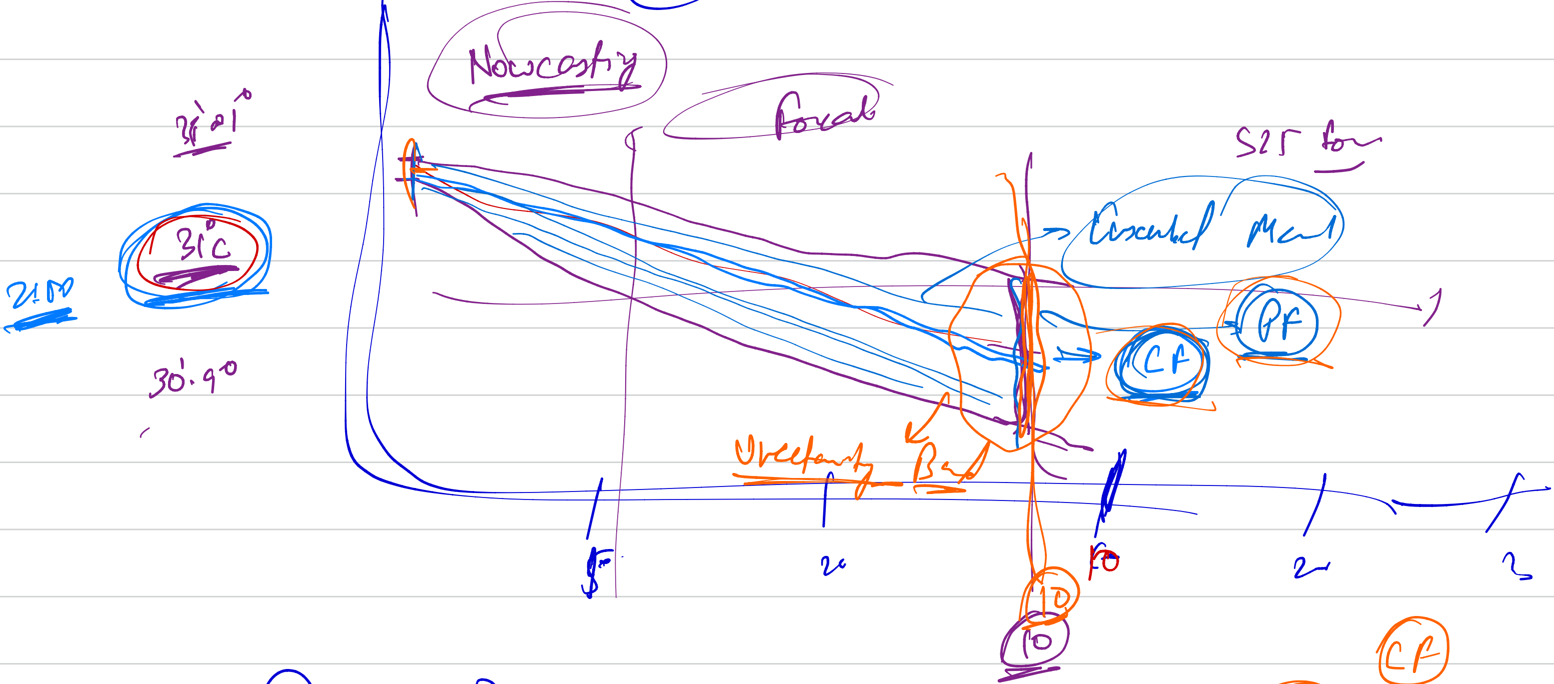
Mode

Model + Obseal → A

③ S2S Model → Better

Pert — A + ○

S2S → Subseasonal - Seasonal



① Physical Model → WRF

Chem

CF

CF+PF / 2

CF → Control Forecast

PF → Perturbed For

Rain

Diff

Cl

IOD

Ahm

Lang

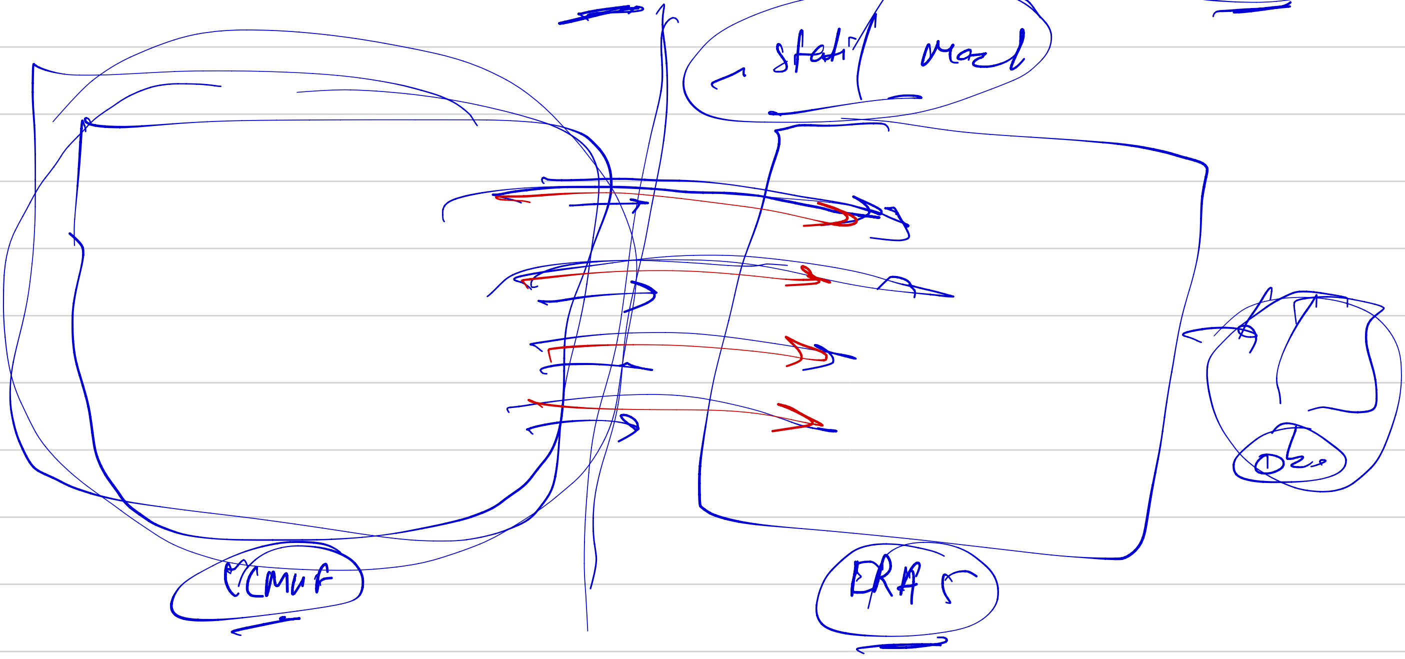
① ECMWF → ~~Ef~~ CYSOR2

↳ Atmospheric + Bu

① Obser → IMD Gridded = .25 x .25

② ERA5 → Rean. = .25 x .25

③ ECMWF → S2S → = 1.25 x 1.25



10 → 40-45

Lead

1 week → Atm

2 → → ① Antecedent, Lan

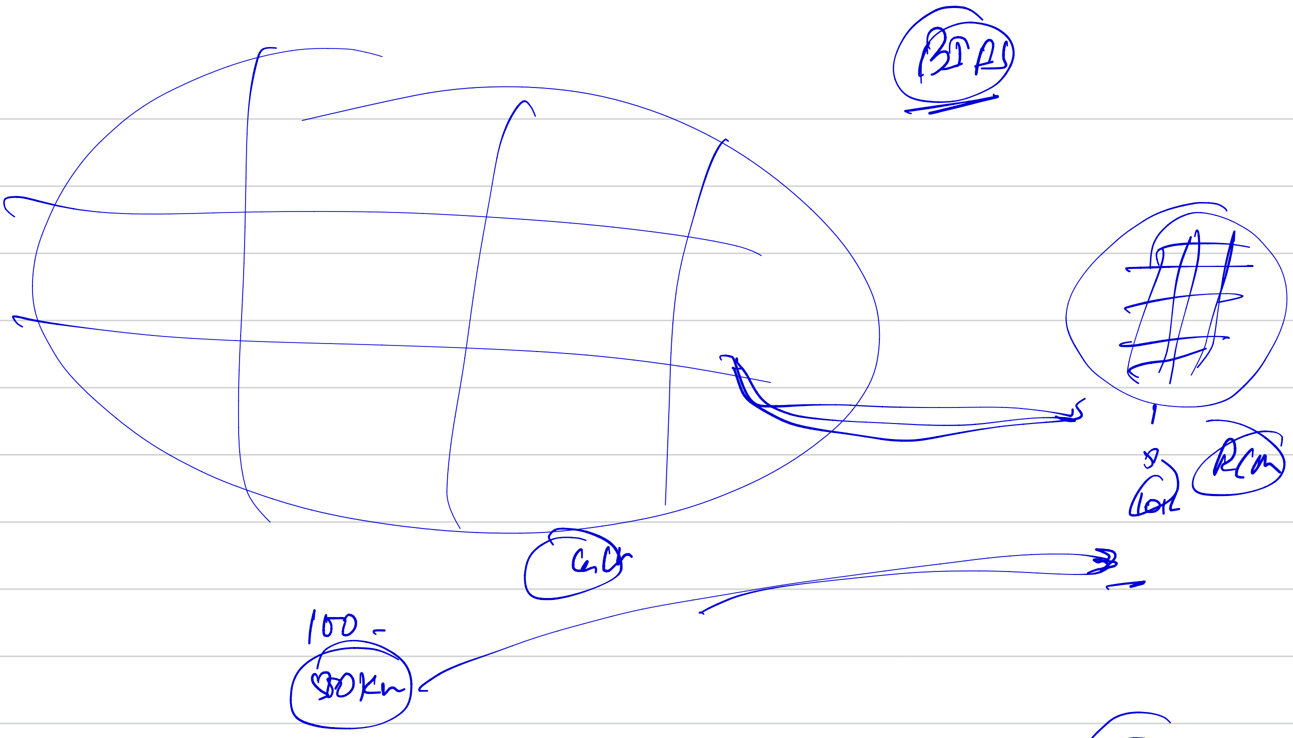
3

4

5 week → Atm

Dee

BERS - Syste Crew



①

Start

Graphcast

Bias Con

Model

Today

Calit

80%

Validat

20%

Hindcast

H2O

2004

2026

Model

SORI

Vless

Hindcast

9 10

20 year

2021

2018

2009

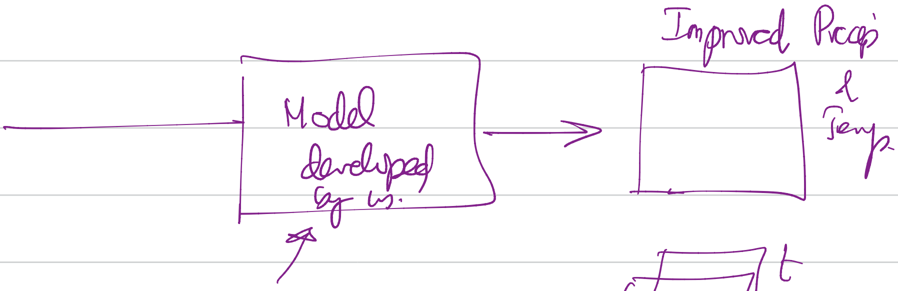
[illegible]

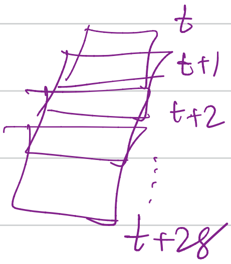
2030

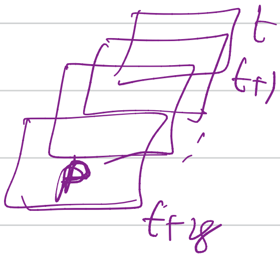
[illegible] – [illegible]

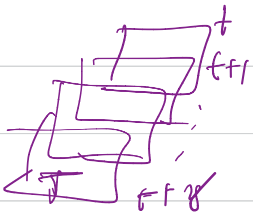
Forecasting. (Appln)

S2S
Model outputs
for next one
month.
(Temp, Pressure,
U, V, RH, ..)
Precip.

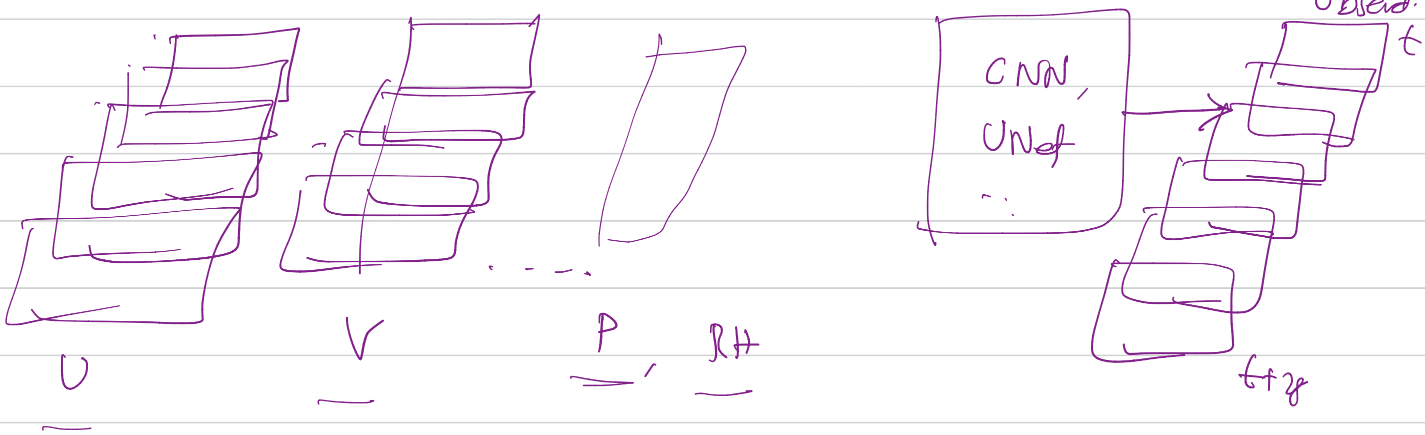








Hindcast (1981- 2024)



1st March 2020 → 1st Mar 2020 → 31st Mar 2020

1st Jan 1981 → 31st Dec 2024 → 31st Jan 1981 to 31th Jan 2025